

Algebra 1: Unit 8, Lesson 2

1. An equation is shown.

$$y = (x + 7)(x + 3)$$

What is the vertex of the graph?

2. What values do the domain and range represent?

3. Functions B , N and S are shown in the table.

Function B	Function N	Function S												
$y = x^2 + 3x - 6$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>-4</td></tr><tr><td>-1</td><td>-1</td></tr><tr><td>0</td><td>2</td></tr><tr><td>1</td><td>5</td></tr><tr><td>2</td><td>8</td></tr></table>	x	y	-2	-4	-1	-1	0	2	1	5	2	8	
x	y													
-2	-4													
-1	-1													
0	2													
1	5													
2	8													

Which of the functions has a domain of $(-\infty, \infty)$?